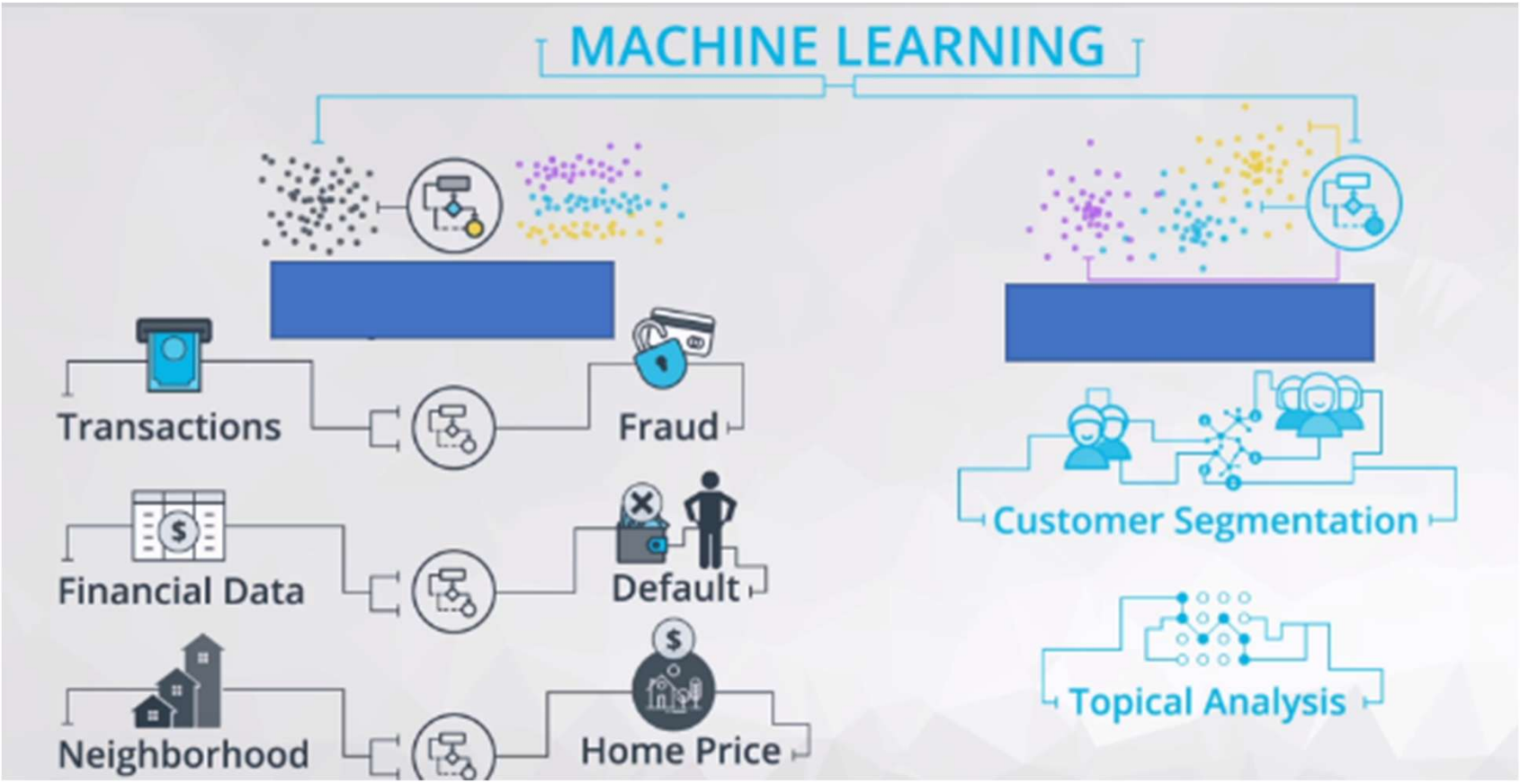
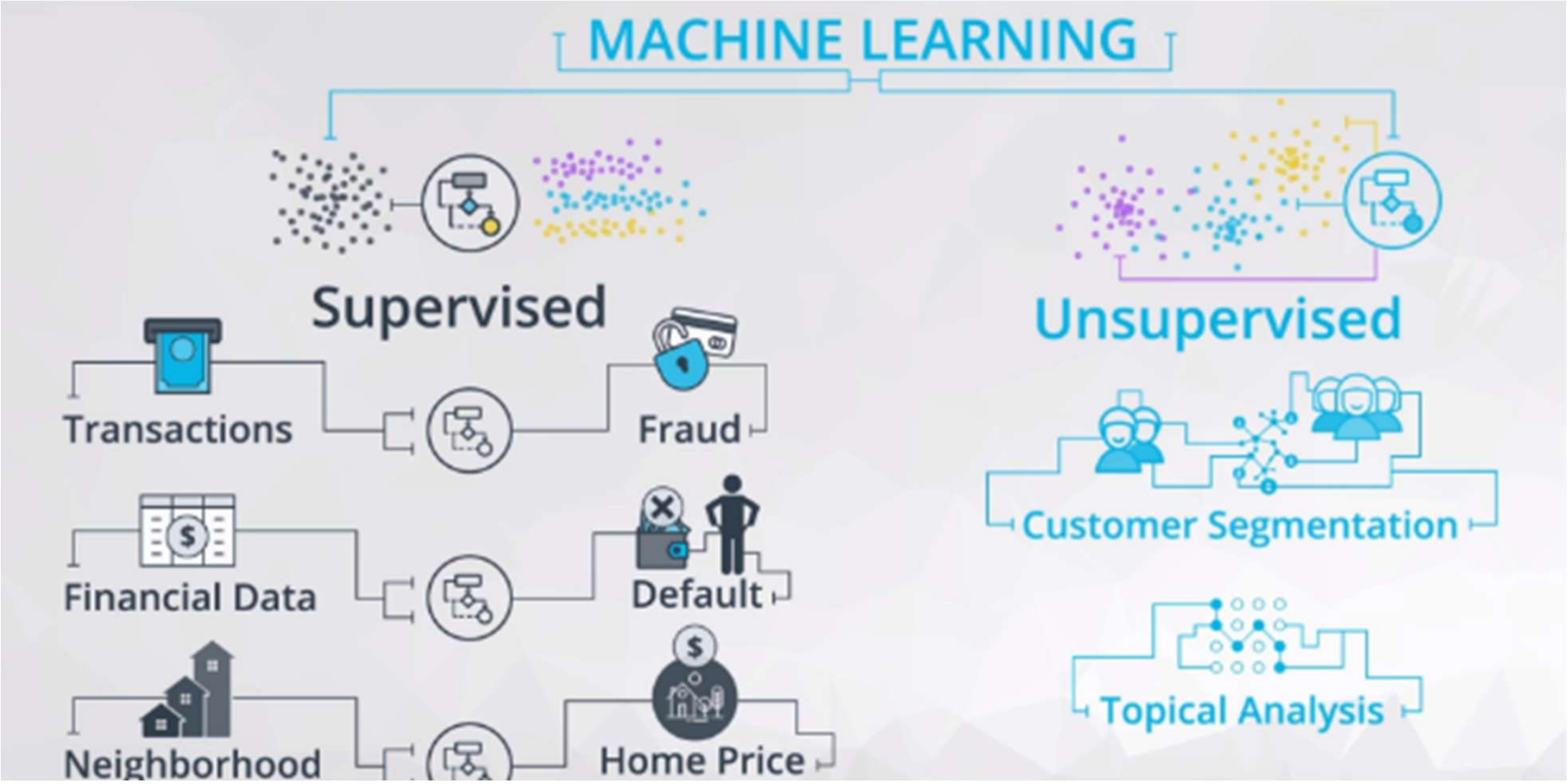
Supervised Machine Learning

Copyright 2025: Jeffrey Saltz, Jeffrey Stanton, and Jasmina Tacheva;
please do not upload.

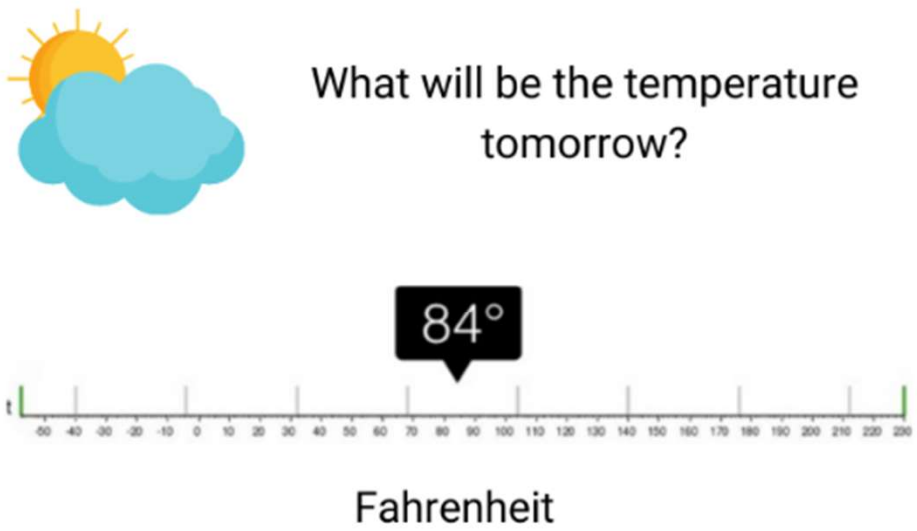
ML Applications

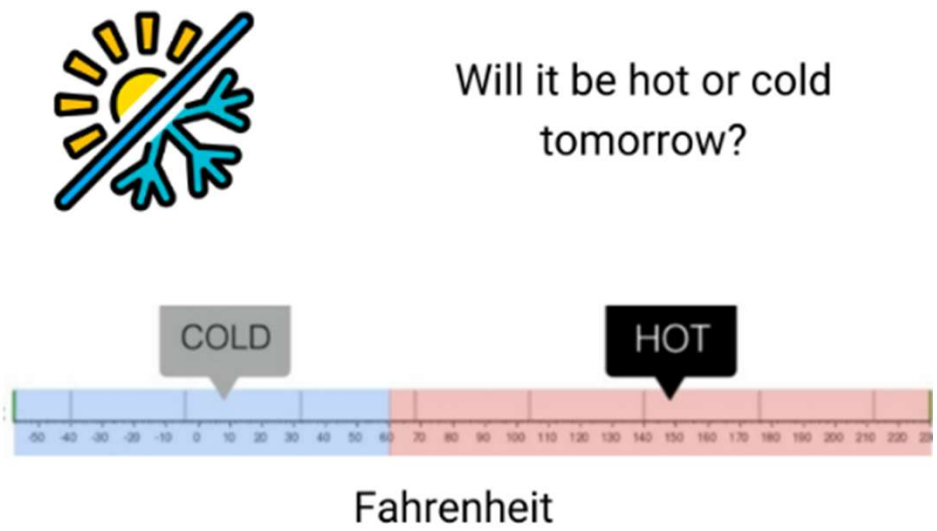


ML Applications

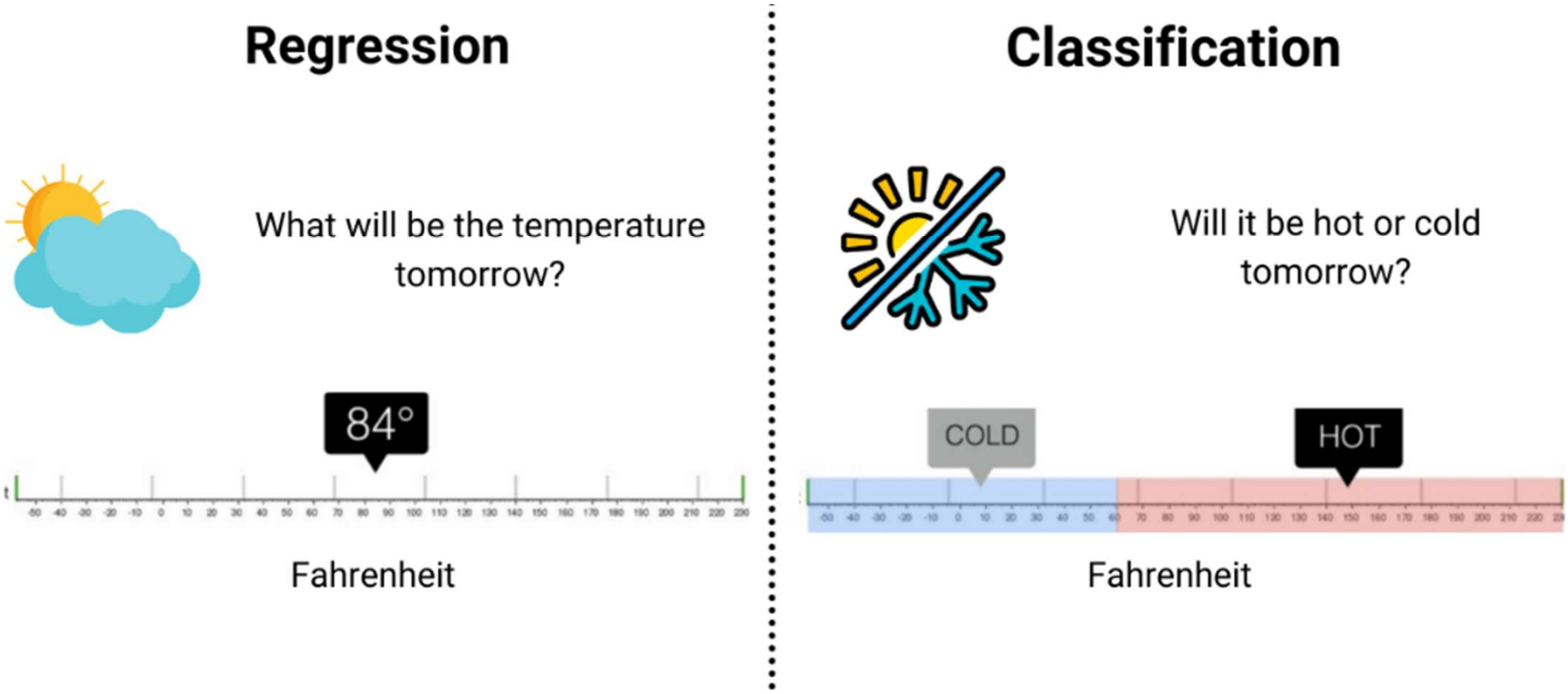


Classification vs. Regression





Classification vs. Regression



Using Classification and Decision Trees

School of Information Studies
SYRACUSE UNIVERSITY

Reviewing The Modeling Process

- Use a substantial number of training cases
  - The machine learning algorithm can use that data to build a model
  - Supervised learning!
- Use the results of this process (i.e., the model) on test data set to determine how well algorithm performed
  - Validate the model on new data
- The result is a model that can be used for prediction
  - Predict data that was not used during training
  - Predict future instances of data

*Note: The model is **not always useful** for explaining results to managers.*

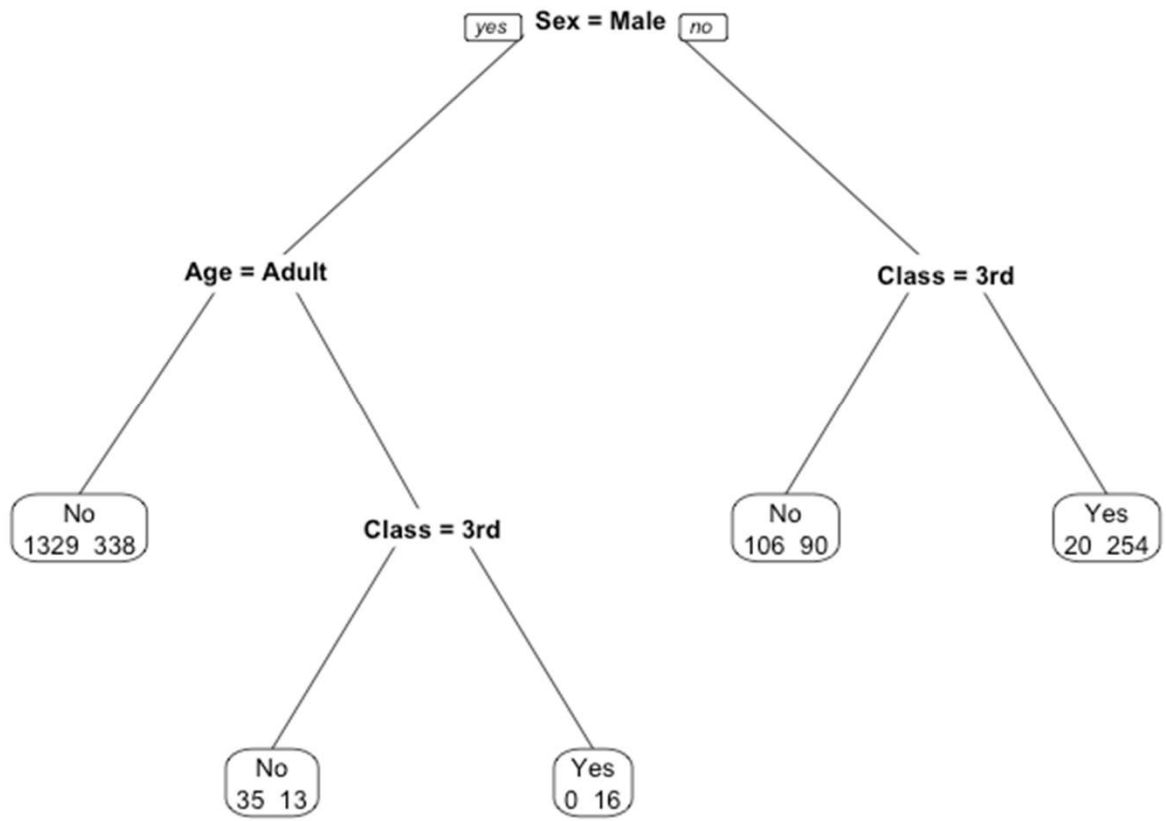
In some cases, algorithms produce results that are not easy to interpret or visualize; for some algorithms there is no output that is like a regression coefficient.

Example of RPART – for Titanic

```
library(rpart)
library(rpart.plot)

url <- "https://intro-datascience.s3.us-east-2.amazonaws.com/titanic.raw.Rdata"

download.file(url, "t.raw.Rdata")
load("t.raw.Rdata")

titanic <- titanic.raw

#Explore the dataset
str(titanic)
'data.frame':	2201 obs. of  4 variables:
 $ Class   : Factor w/ 4 levels "1st","2nd","3rd",..: 3 3 3 3 3 3 3 3 3 3 ...
 $ Sex     : Factor w/ 2 levels "Female","Male": 2 2 2 2 2 2 2 2 2 2 ...
 $ Age     : Factor w/ 2 levels "Adult","Child": 2 2 2 2 2 2 2 2 2 2 ...
 $ Survived: Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 1 1 1 1
```

Example of RPART – for Titanic

```
#Build the model
cartTree <- rpart(Survived ~ ., data = titanic)
prp(cartTree, faclen = 0. cex = 0.8. extra = 1)
```



For left pointing leaves:

- The first number: # of correctly classified cases
- The second number: # incorrectly classified.

For right pointing leaves, vice versa.

Predicting values using the Model

```
predictValues <- predict(cartTree, newdata=titanic,
                                                    type = "class")

titanic[1,]
    Class   Sex    Age     Survived
3   3rd   Male   Child       No

predictValues[1]
1
No
Levels: No Yes
```

What was the Accuracy

```
actualSurvived <- as.factor(titanic$Survived == "Yes")
confMatrix <- table(predictedSurvived,actualSurvived)
confMatrix

                        actualSurvived
predictedSurvived   FALSE   TRUE
        FALSE        1470    441
        TRUE           20    270

accuracy <- 1 - (sum(confMatrix)  -  sum(diag(confMatrix)))
                                                / sum(confMatrix)
accuracy
0.7905498
```

Questions

Is 79% good?

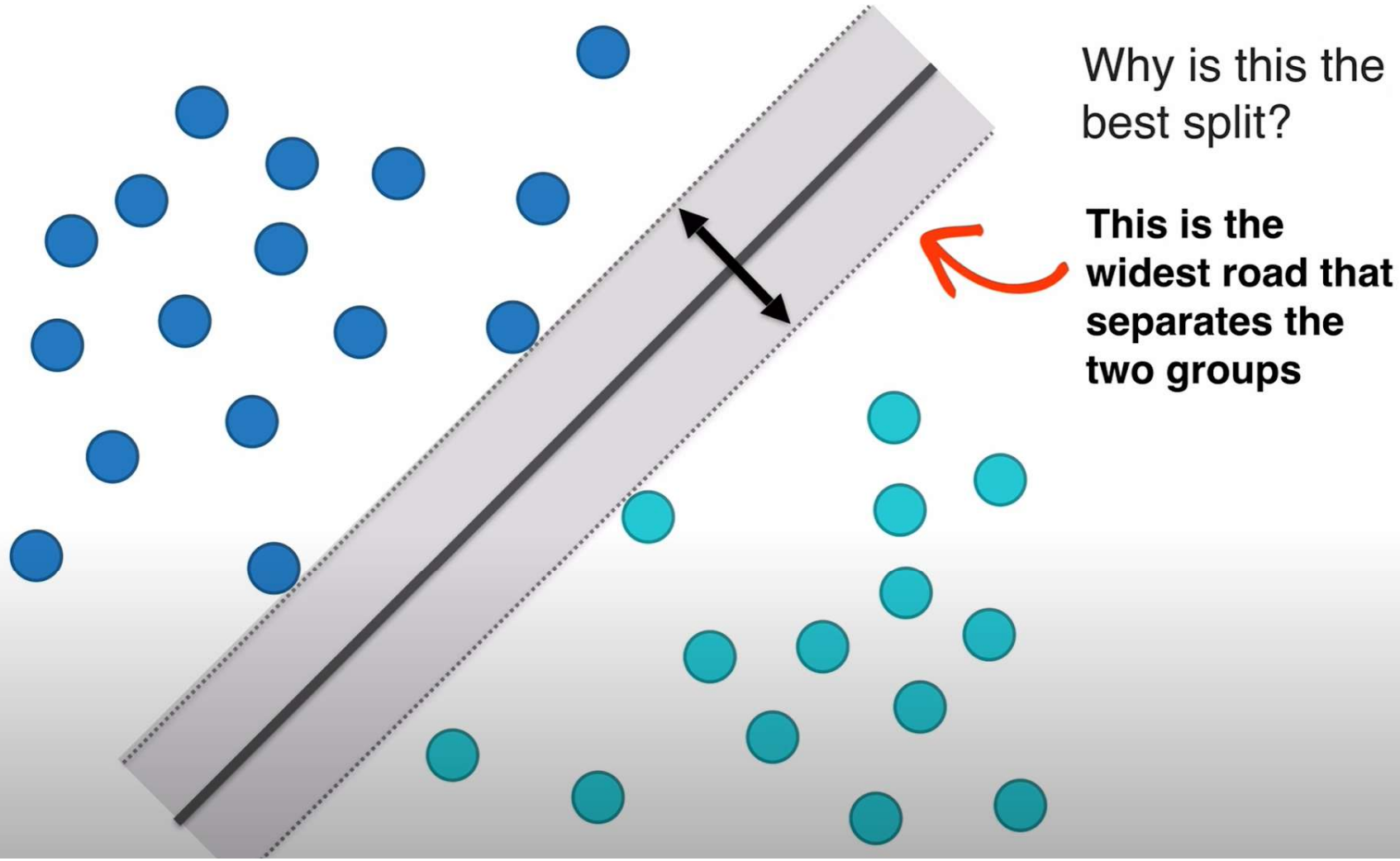
What is overfitting?

How might that be relevant?

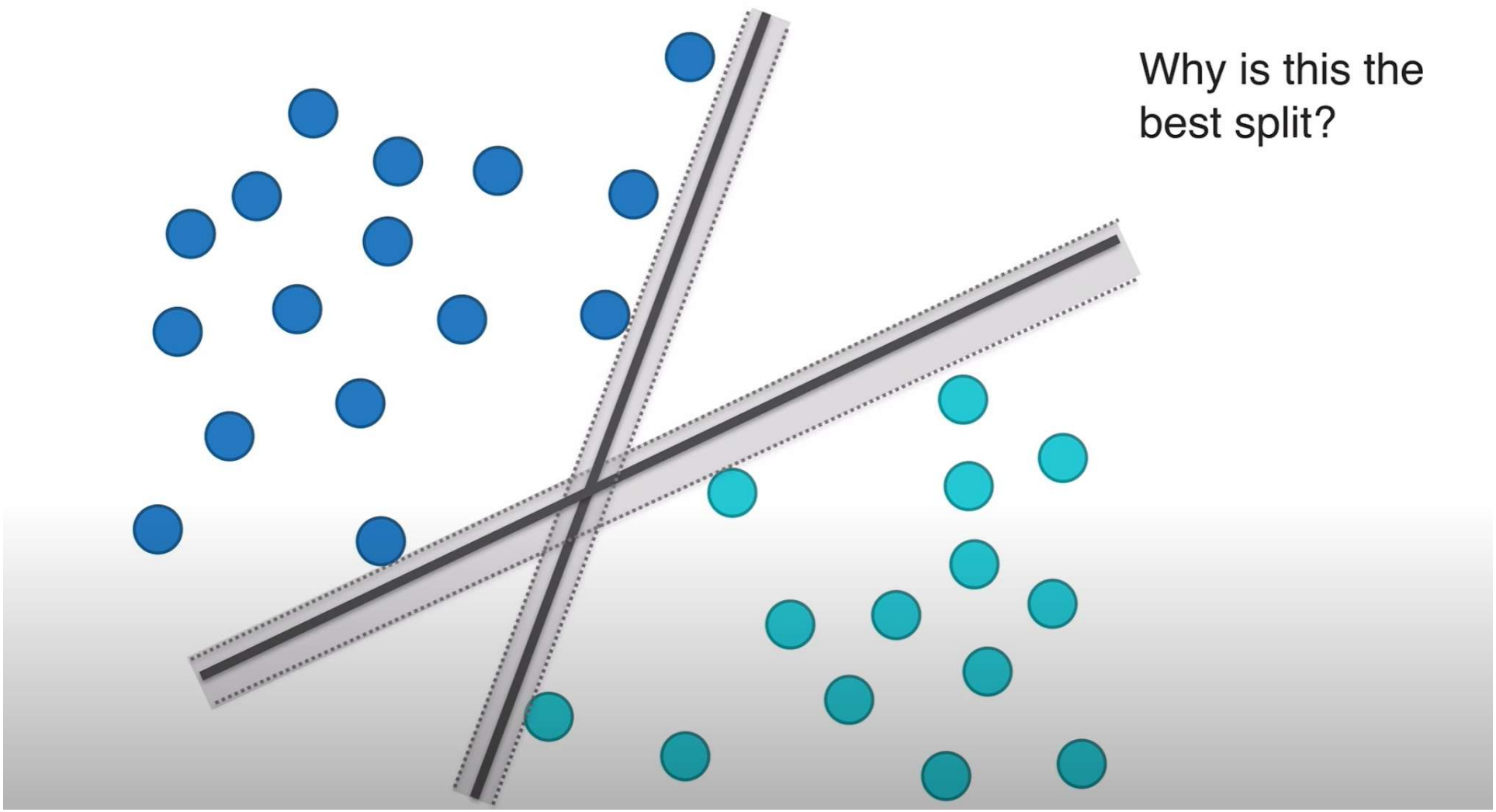
Using
Support Vector Machines

School of Information Studies
SYRACUSE UNIVERSITY

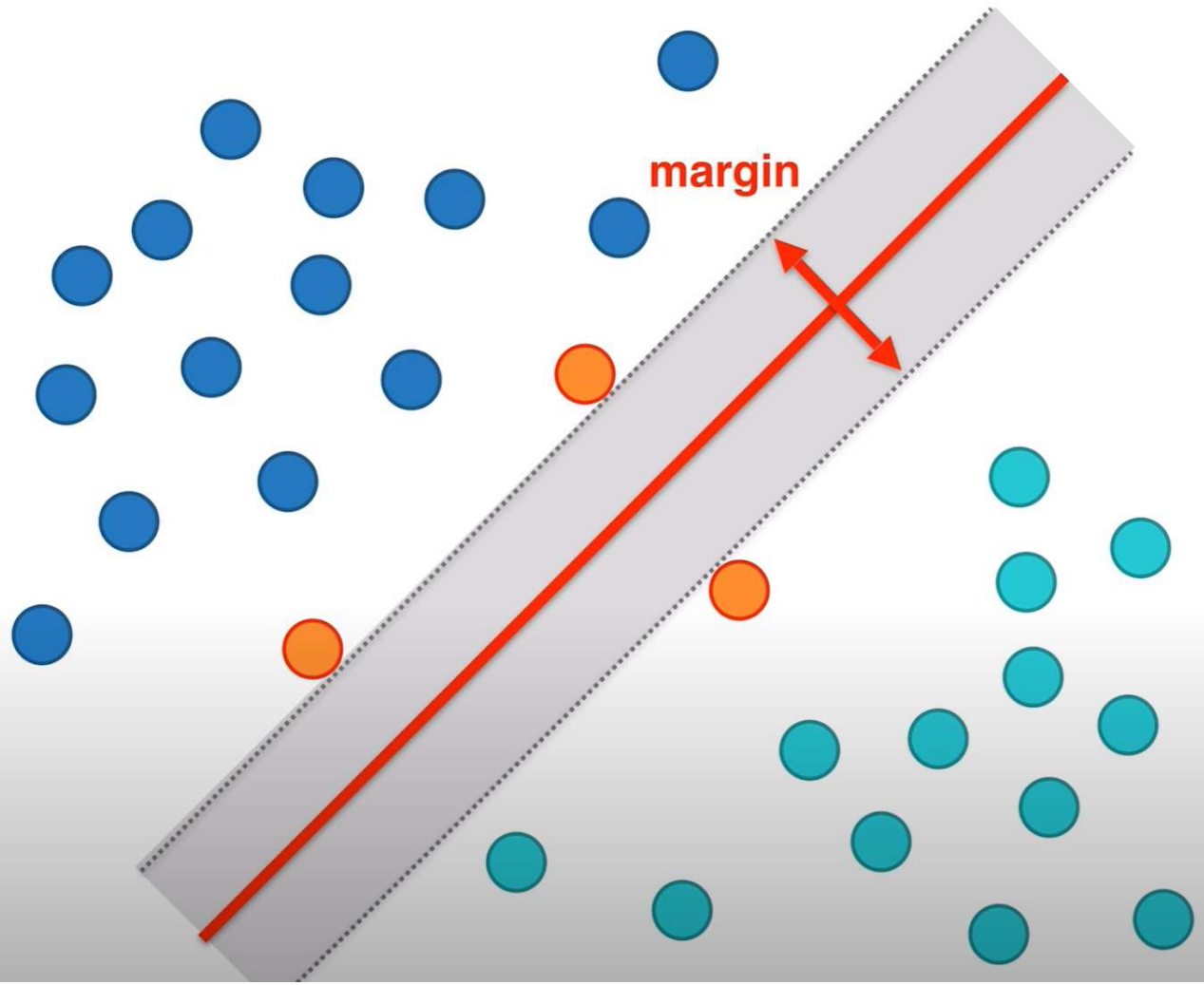
SVM


Why is this the best split?
This is the widest road that separates the two groups

SVM



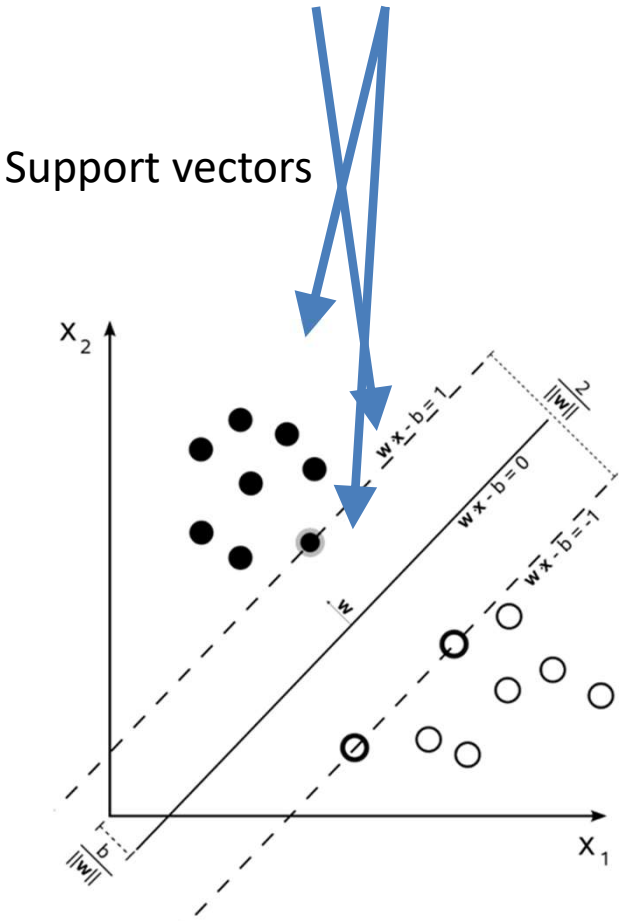
SVM



Why is this the best split?

The distance between the support vectors and the hyperplane are as far as possible

SVM: What are support vectors?



- Very **different approach to prediction** compared with the linear methods and tree methods
- Based on **statistical learning theory**
- Use the **cases that are difficult to predict** to help define a robust prediction model (because any model can handle "easy" cases!)
- Support vectors ***are* the harder-to-predict cases** on a defined border + margin that separate the cases
- The **profiles of predictor scores** define the **support** or border + margin that serves to classify future cases

Support vectors

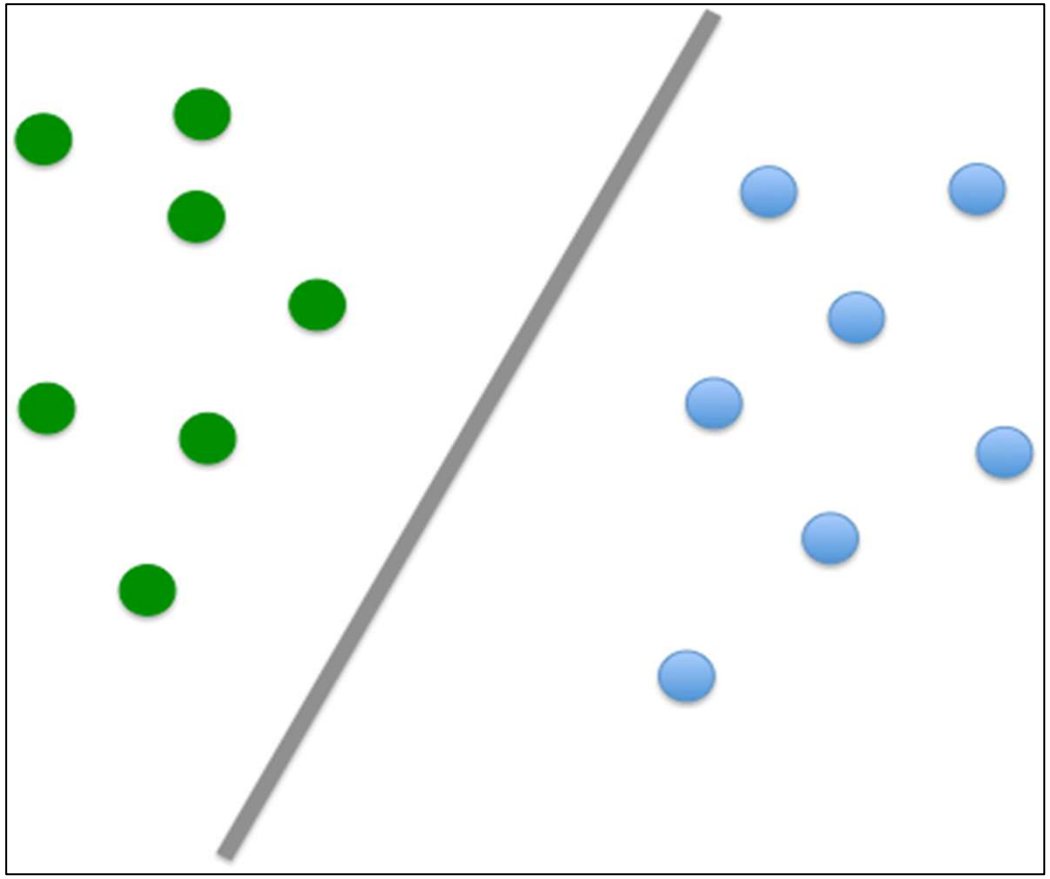
- The data points that are most difficult to classify, e.g. a cat groomed like a dog!

Support vectors

- The data points that are most difficult to classify, e.g. a cat groomed like a dog!

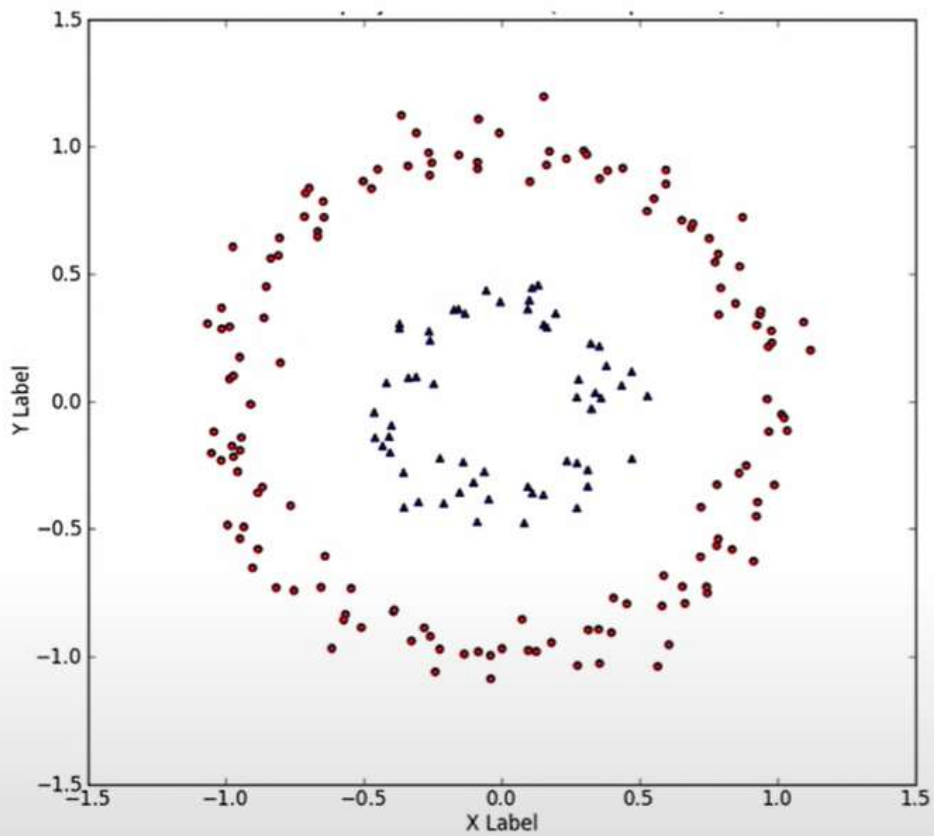
Visualizing the Algorithm

This is Easy



Visualizing the Algorithm

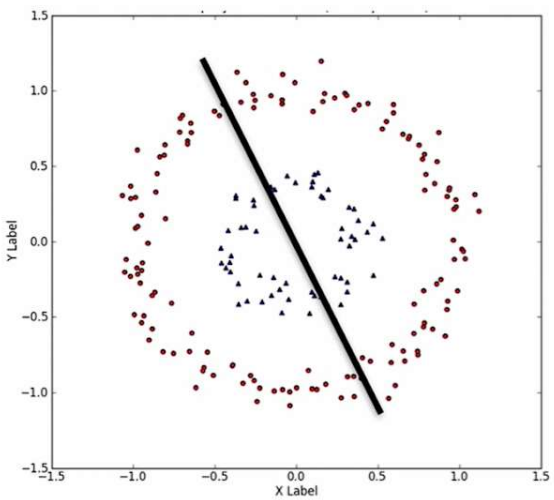
What About This?



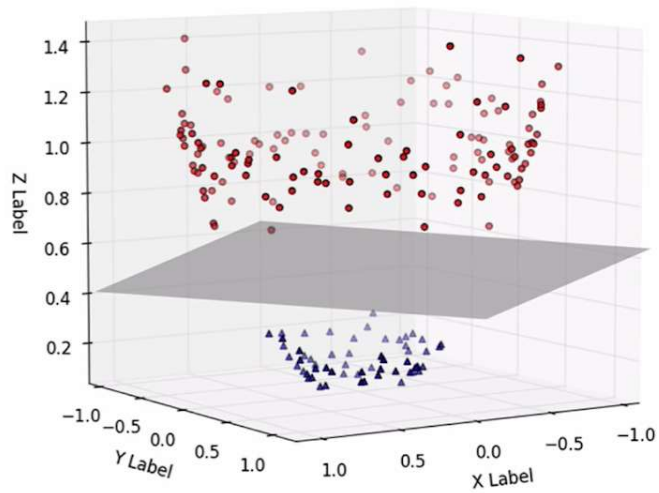
Visualizing the Algorithm: The Kernel Trick

2D to 3D Mapping

Kernel Trick: Visual



Kernel

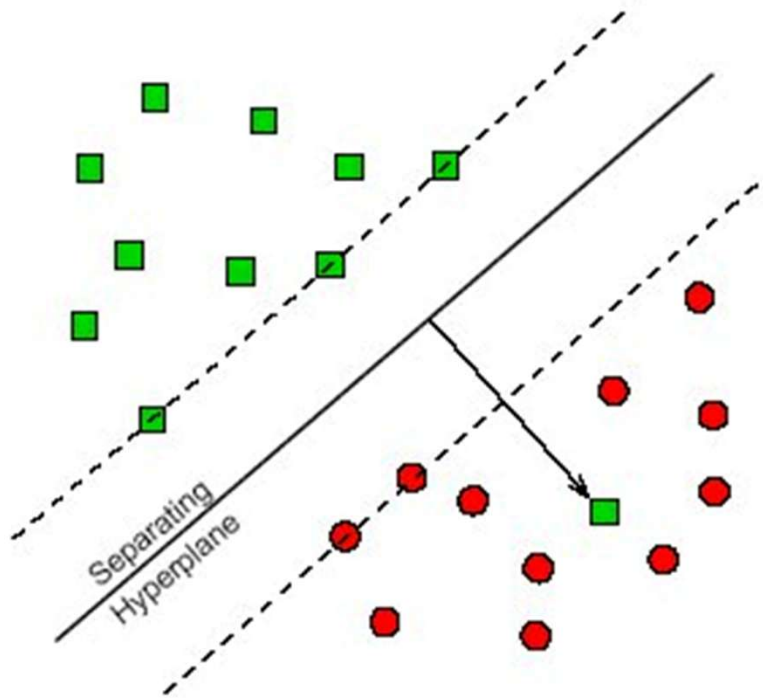


Kernel Options
- Linear
- Radial Basis Function
- Polynomial
- Sigmoid

Why Error Might Not Be Bad

Non-separable training sets

Use linear separation, but admit training errors.

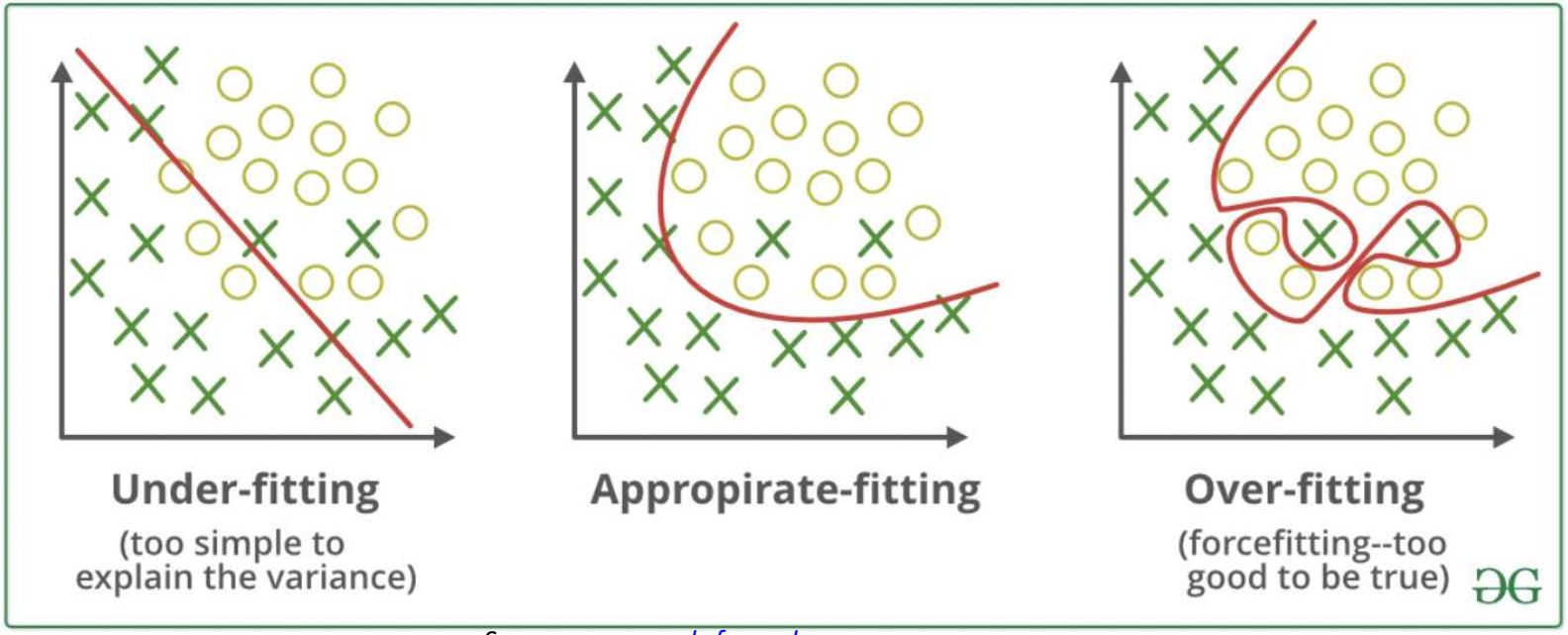


Penalty of error: distance to hyperplane multiplied by *error cost* C.

Over and Underfitting

- **Overfitting**: a model is trained so accurately that it learns the noise in the data. *Models have trouble predicting new data based on the training data because this noise cannot be applied to new data.*

- **Underfitting**: a model doesn't capture the underlying insights of the data. *Models have trouble predicting new data because the model is not accurate enough – the model is too simple*



Source: www.geeksforgeeks.org

Cost Parameter ('C')

ksvm Cost Parameter Impact Summary

Higher Cost 'C' Value	**Fewer Classification Mistakes** **Fewer Problem Points**	**Smaller Margin of Separation**	**Specialized Model**	**Higher Cross-Validation Error**	**Lower Training Error**
Lower Cost 'C' Value	**More Classification Mistakes** **More Problem Points**	**Higher Margin of Separation**	**Generalized Model**	**Lower Cross-Validation Error**	**Higher Training Error**

Use SVM to Explore the Credit Data

```
#Step 1: load the required packages was already done

#Step 2: Create Training and Test sets
data("GermanCredit")
subCredit <- GermanCredit[,1:10]

# makes the sampling predictable
set.seed(111)

# Randomly sample elements to go into a training data set
trainList <- createDataPartition(y=subCredit$Class,p=.80,list=FALSE)
trainSet <- subCredit[trainList,]
testSet <- subCredit[-trainList,]
```

Use SVM to Explore the Credit Data

```
#Train the SVM model
fit2 <- train(Class ~ ., data=trainSet, method="svmRadial",
                                        preProc=c("center","scale"))

#Assess Fit with new data
predOut <- predict(fit2, newdata=testSet)
```

Explore the confusion matrix

```
# Review the error – use the built in 'confusionMatrix'
confusion <- confusionMatrix(predOut, testSet$Class)
confusion
Confusion Matrix and Statistics

          Reference
Prediction Bad Good
       Bad  10    3
      Good  50  137

               Accuracy : 0.735
                 95% CI : (0.6681, 0.7948)
    No Information Rate : 0.7
    P-Value [Acc > NIR] : 0.1579
```

Explore the “C” parameter

fit2

Support Vector Machines with Radial Basis Function Kernel

800 samples

9 predictor

2 classes: 'Bad', 'Good'

Pre-processing: centered (9), scaled (9)

Resampling: Bootstrapped (25 reps)

Summary of sample sizes: 800, 800, 800, 800, 800, 800, ...

Resampling results across tuning parameters:

C	Accuracy
0.25	0.6969592
0.50	0.6974949
1.00	0.6989207

Weekly Quiz

In Blackboard

Time Limit:
4 Minutes